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| **Status** | Finished |
| **Started** | Wednesday, 21 October 2026, 10:08 PM |
| **Completed** | Wednesday, 21 October 2026, 10:14 PM |
| **Duration** | 6 mins 2 secs |
| **Grade** | **14.00** out of 15.00 (**93.33**%) |

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. There must be a mechanism to delete and/or insert elements in the list ✓

b. Number values should only be present

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c. There should be direct access to the middle element in any sublist

d. The list must be sorted

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search requires the list to be sorted.

b. Linear search is more efficient than binary search on large lists.

c. Linear search divides the list into halves.

d. Linear search can be applied to both sorted and unsorted lists. ✓

Question **3**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It always starts from the beginning of the list

b. It can be applied only if the list is sorted ✓

c. It compares elements sequentially

d. It works on unsorted lists

Question **4**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 90 and 100

b. 90 and 99 ✓

c. 89 and 94

d. 94 and 99

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

a. Merge search

b. Linear search ✓

c. Bubble search

d. Quick search

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

a. Hashing

b. Linear search ✓

c. Dynamic search

d. Binary search

Question **7**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The middle element is chosen randomly

b. The search stops

c. The lower middle element is chosen as the middle element ✗

d. The higher middle element is chosen as the middle element

Question **8**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By comparing each element sequentially
b. By dividing the list length by two ✓
c. By using a hash function
d. By starting from the first element

Question **9**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Binary search
b. Hash search
c. Linear search ✓
d. Both (1) & (3)

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is sorted ✓
b. When the list contains duplicate elements
c. When the list is very small
d. When the list is unsorted

Question **11**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n ✓
b. n-1
c. log n
d. n/2

Question **12**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. 1
b. log n
c. n ✓
d. n/2

Question **13**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(1)
b. O(n log n)
c. O(log n) ✓
d. O👎

Question **14**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Finding
b. Searching ✓
c. Discovering
d. Mining

Question **15**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the entire list
b. In the right sublist
c. At the beginning of the list
d. In the left sublist ✓